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A Comparative Systematic Review of Convolutional Neural Networks (CNN) versus Multilayer Perceptrons (MLP) in Image Classification

Introduction to Artificial Intelligence – ARTI 401 – 7MA2

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1. Abstract

Image classification is still a big challenge for AI today. For a long time, the Multilayer Perceptron (MLP) was the main algorithm for pattern recognition. However, as systems started to manage images in high dimensional visual pixels, the traditional MLP structure could not keep up with the accuracy and speed needed. This led to the rise of Convolutional Neural Networks (CNNs), which is inspired by the human brain, processing visual information. This review compares and evaluates CNNs and MLPs in image classification based on accuracy, computational efficiency, and reliability across complex datasets, and the usefulness of combining both models together (Hybrid model).

The results show a clear difference in performance between the two algorithms. On the Chars74k dataset, CNNs got an accuracy of 88.39%, while MLPs only 43.4%. This is also true for the Fashion-MNIST dataset, where CNNs outperformed MLPs by 3.5%. In the data mining dataset, CNNs performed better too, with an R^2 of 0.8472 compared to MLP 0.8286. However, MLPs struggle with overfitting, which means they do not always work well with complex and large data. Recent changes like "Linear Bottleneck Layers" have helped MLPs reach 78.6% accuracy at CIFAR-10, showing they might still be useful in low resource images. So, this review shows that hybrid models, which mix MLPs and CNNs, perform better in some areas. For example, models used for diagnosing diabetes and COVID-19 have achieved over 96% accuracy, which is much better than using just one algorithm. So, while CNNs are the best for image classification as a standalone model because of their ability to find spatial features, the future is hybrid models that use both MLPs and CNNs.

Keywords: Convolutional Neural Networks (CNN); Multilayer Perceptron (MLP); Deep Learning; Image Classification; Hybrid Models; Medical Diagnostics; Fashion-MNIST; Chars74k; Educational Data Mining.

2. Introduction

2.1 Historical Context and Background

In the last two decades, the landscape of Artificial Intelligence has changed such that Machine Learning has been replaced by Deep Learning as the go to techniques. This is due to larger data sets, more complicated structures, especially in fields that desire higher resolution characteristics. For example, in the medical field of pediatric epilepsy detection, conventional techniques do not note the important subtleties that make deep learning techniques even more applicable. In computer vision, however, two types of neural networks come to the forefront, one of which is distinctly different from the other: multilayer perception (MLP) and Convolutional Neural Network (CNN).

MLP, or Multi-Layer Perceptron, is a type of fully-connected neural network that consists of an input layer, multiple hidden layers, and an output layer through which each neuron is connected to each neuron in the subsequent layer (fully connected algorithm). This works well with detection issues characterized by non-linear solutions, but when it comes to computer vision, it does not work. For example, in computer vision, MLP needs to flatten the image sent to it into one layer of numbers. Yet, this vulnerability can be corrected with Linear Bottleneck Layers, showing that MLPs are still viable where appropriate.

CNNs were created for image processing with larger pixels. They came into the spotlight when AlexNet was introduced into ImageNet and reduced the error rate from 26% to 15.3%. CNNs leverage convolution, which is a mathematical operation used to analyze parts of an image with certain filters through which it can discern gradients and shapes without losing spatial characteristics. Therefore, CNN is seen as the best solution for anything from facial recognition to remote sensing.

2.2 Visualizing Data Complexity

To understand why we need these advanced models, we need to see how complicated image datasets are. For example, digits (standard MNIST) are easy to classify, but modern datasets like Fashion-MNIST contain complicated textures and combined visuals.

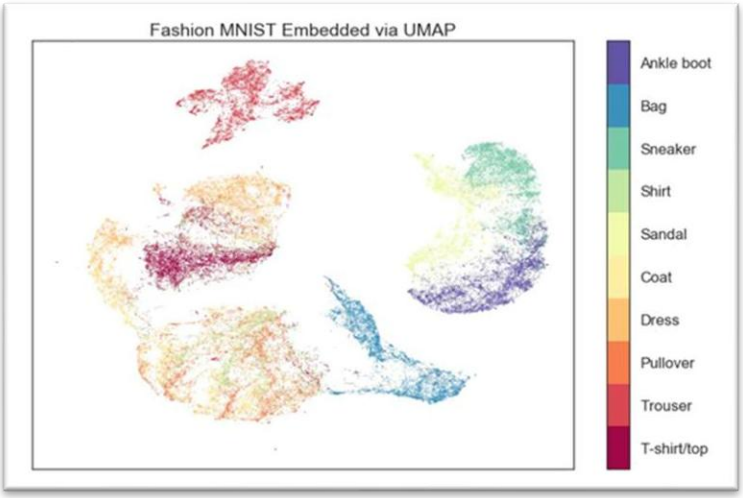

Figure 1: *UMAP visualization illustrating the high complexity and clustering of the Fashion-MNIST dataset, clearing why simple linear models often fail [2].*

2.3 Rationale and Research Question

While CNN is the more researched option in academia, MLP has seen more use in industry due to greater facilitation and ease of use and new architecture such as Vision Transformers (ViT) may make CNN not the end-all-be-all solution for the future. Thus, this paper will attempt to answer the following question:

- *"How does the classification accuracy, computational robustness, and training efficiency of Convolutional Neural Networks (CNN) compared to Multilayer Perceptrons (MLP) when applied to standard, diverse image classification datasets?"*

Methodology

3.1 Search Protocol

To ensure that this paper is both transparent and trustworthy by future researchers, a planned search strategies was made. The search was formally made on November 20, 2025.

The primary sources for literature and references were IEEE Xplore, Google Scholar, and the IAU Digital Library. The databases that have been selected have the highest impact factor journals in the field of Deep Learning and Artificial Intelligence.

3.2 Search Strategy and Terms

To assure the most applicable results, a Boolean search string was compiled. This method combined search terms associated with the specific algorithm being studied with the image processing task. The following search string was compiled:

("Convolutional Neural Network" OR "CNN") **AND** ("Multilayer Perceptron" OR "MLP" OR "FCNN") **AND** ("Image Classification" OR "Character Recognition")

3.3 Inclusion and Exclusion Criteria

The search results were filtered based on a simple set of criteria to ensure quality and relevance:

- **Inclusion Criteria:**
  - **Date Range:** Papers published between **2010 and 2025** were selected to ensure the review reflects the current state of the art.
  - **Datasets:** Studies that used standard, reproducible image benchmarks (MNIST, Fashion-MNIST, Chars74k, ImageNet) as well as real world datasets (clinical and educational data) for hybrid model evaluation.
  - **Language:** Only papers written in English were chosen.
- **Exclusion Criteria:**
  - Papers that did not discuss, mention, or compare any of the Neural Network architectures.
  - Duplicate records or short abstracts lacking details.

3.4 PRISMA Flow Diagram

The study selection process followed the **PRISMA** guidelines. This process involved identification, screening, eligibility, and inclusion phases.

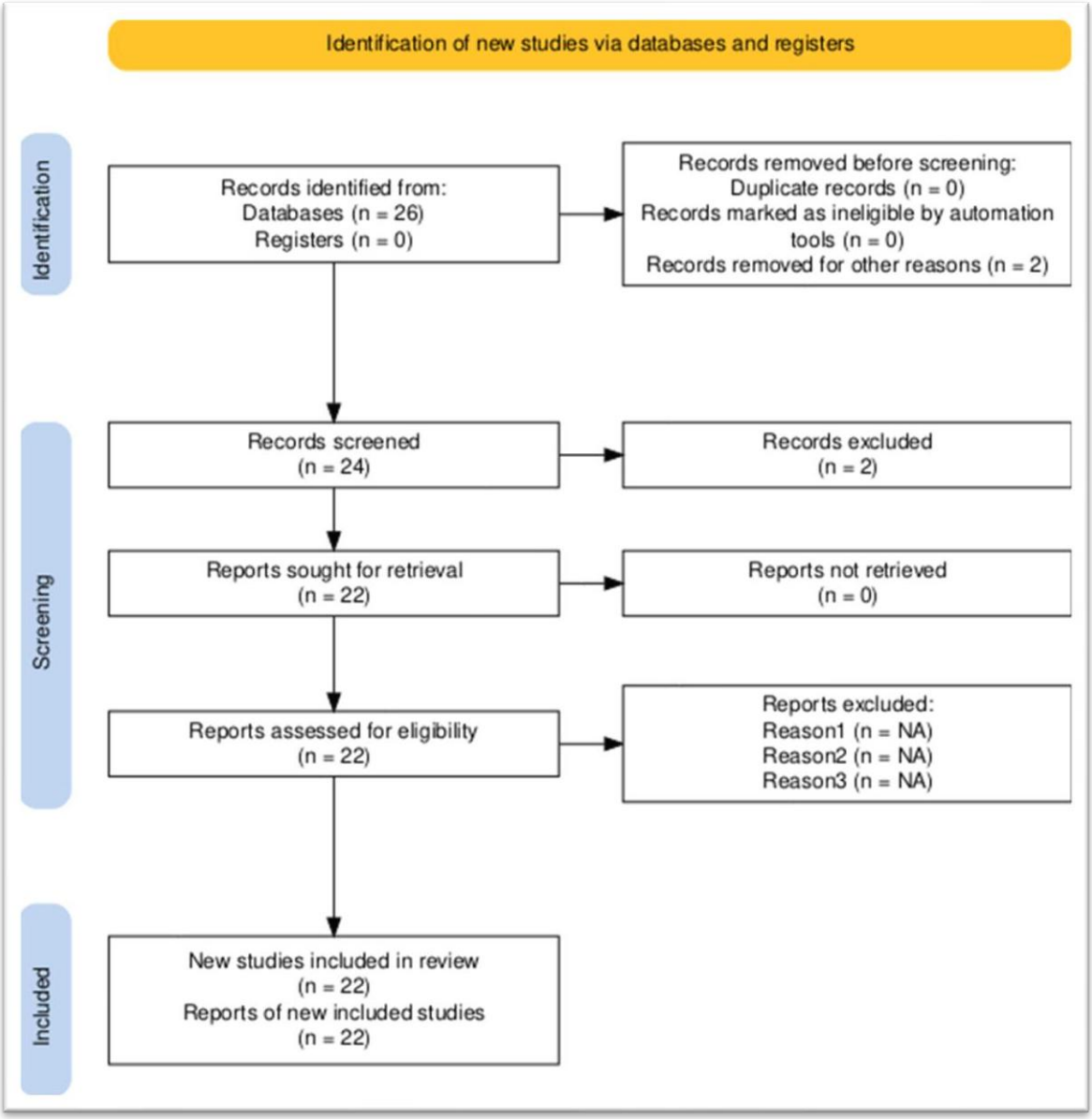

Figure 2: *PRISMA flow diagram illustrating the selection process of the research papers used in this paper.*

4. Results – Study Selection & Characteristics

Twenty two papers were selected as references. Below is a detailed characterization of the key studies, categorized by their primary contribution:

A. Foundational Comparisons & Theory

1. **Ahmed et al. (2022) [1]:** Reviews ML vs. DL in medical contexts.
2. **Damle (2021) [2]:** Experimental of MLP vs. CNN on Fashion-MNIST.
3. **Purwono et al. (2022) [3]:** breakdown of CNN layers.
4. **Zhao et al. (2021) [4]:** Large scale study on ImageNet (CNN vs. MLP vs. Transformer).
5. **Krizhevsky et al. (2012) [19]:** AlexNet paper demonstrating the first major win of Deep CNN over traditional methods.

B. Image & Character Classification

6. **Ben Driss et al. (2017) [5]:** Comparison on the Chars74k dataset (natural images).
7. **Baldominos et al. (2019) [6]:** Survey of MNIST results.
8. **Airlangga (2024) [11]:** Deep MLP vs CNN vs LSTM for predicting student performance, finding CNNs superior.
9. **Hartpence & Kwasinski (2021) [14]:** Explored ensembles of CNN and MLP for network packet classification, finding that CNNs excel at UDP traffic while MLPs are better for TCP traffic.

C. Hybrid & Advanced Models (2024-2025)

10. **Ahsan et al. (2020) [7]:** made a Hybrid MLP-CNN model for COVID-19.
11. **Nizamuddin et al. (2024) [12]:** Developed a hybrid MLP-CNN model for chronic kidney disease, utilizing health monitoring.
12. **Leng et al. (2024) [15]:** Used an Exception-MLP architecture to predict anti-VEGF therapy outcomes, demonstrating that combining image data (OCT) with data (MLP) yields better regression results.
13. **Fan (2025) [16]:** Proposed a massive ensemble of CNN, LSTM, and MLP for diabetes, achieving 98.28% accuracy.
14. **Nguyen et al. (2025) [13]:** Applied CNN and MLP models to health monitoring of bridges, proving the usefulness of these architectures beyond healthcare.

D. Optimization & Future Directions

15. **Maurício et al. (2023) [8]:** Compares CNNs with Vision Transformers (ViT).
16. **Mienye et al. (2024) [9]:** A 2024 review of CNN evolution.
17. **Lin et al. (2016) [10]:** Investigates "Linear Bottleneck Layers" to improve MLPs.
18. **Ioannou et al. (2017) [17]:** Discusses "Deep Roots" and hierarchical filter groups to improve CNN efficiency.
19. **Feinman & Lake (2018) [18]**: Analyzes biases, showing that simple neural networks can learn shape biases like humans.
20. • **Ma & Cui (2025) [20]:** A comparative analysis of FCNN and CNN architectures, demonstrating efficiency in noise frequency handling.
21. • **Kang & Wang (2014) [21]:** Proposes fully convolutional neural networks (FCN) as an architectural evolution for dense prediction tasks like crowd segmentation.
22. • **Cui et al. (2021) [22]:** Investigates multi-feature fusion decision strategies to enhance CNN classification performance in complex SAR imagery.

The 8 papers listed in **Table 1** are the Experimental/Primary studies. The other 11 papers are mostly Reviews.

Reference	Year	Dataset Used	Focus Area
Ahmed et al. [1]	2022	Medical EEG (CHB-MIT)	ML vs. DL in Pediatric Epilepsy (Review)
Damle [2]	2021	Fashion-MNIST	CNN vs. MLP Hyperparameter Optimization
Purwono et al. [3]	2022	N/A (Theory)	CNN Layer Architecture (Theoretical Review)
Zhao et al. [4]	2021	ImageNet-1K	Empirical Study of CNN, MLP, and Transformer
Ben Driss et al. [5]	2017	Chars74k	Natural Character Recognition (MLP vs. CNN)
Baldominos et al. [6]	2019	MNIST / EMNIST	Survey of Handwritten Character Recognition
Ahsan et al. [7]	2020	COVID-19 X-ray	Hybrid MLP-CNN Model for Diagnosis
Maurício et al. [8]	2023	ImageNet / Various	Vision Transformers vs. CNN (Review)
Mienye et al. [9]	2024	Multiple (Review)	Comprehensive Review of Deep CNNs
Lin et al. [10]	2016	CIFAR-10 / MNIST	Improving MLPs with Linear Bottlenecks
Airlangga [11]	2024	Student Performance Data	Educational Prediction (MLP vs. CNN vs. LSTM)
Nizamuddin et al. [12]	2024	Clinical Data	Real-time Health Monitoring (Hybrid)
Nguyen et al. [13]	2025	Bridge Vibration Signals	Structural Health Monitoring (CNN + MLP)
Hartpence et al. [14]	2021	Network Traffic Data	Packet Classification Ensembles (Security)
Leng et al. [15]	2024	OCT Images / Clinical Data	Diabetic Macular Edema (Hybrid Xception-MLP)
Fan [16]	2025	Diabetes Clinical Data	Hybrid CNN-LSTM-MLP Ensemble
Ioannou et al. [17]	2017	CIFAR-10 / ImageNet	CNN Efficiency (Deep Roots / Filter Groups)
Feinman & Lake [18]	2018	Synthetic Shapes	Inductive Biases in Neural Networks
Krizhevsky et al. [19]	2012	ImageNet	Foundational Deep CNN (AlexNet)
Ma & Cui [20]	2025	Speech/Noise Frequencies	FCNN vs CNN for Speech Denoising
Kang & Wang [21]	2014	Crowd Datasets	Fully CNN for Crowd Segmentation
Cui et al. [22]	2021	SAR Images	Multi-feature Fusion Decision CNN

Table 1: *Summary of characteristics of key experimental studies included.*

5. Critical Appraisal and Architectural Analysis

5.1 Architectural Contrast: CNN vs. MLP

The significant performance differences across these two algorithms are based on architecture and specifically how they respond to the spatial features found in images.

Multilayer Perceptron (MLP): MLP is a fully connected one, but as an image recognizer, it's not useful since it needs the 2D pixel array to be transformed into a 1D array [6][7]. Thus, it removes any sense of pixel proximity to where the network needs to learn features across the image rather than in a localized section, which seems impossible and tedious. In addition, Damle states [2] that this is a highly interconnected and complex process where every neuron can connect to every other neuron in the layer after that, resulting in difficulties such as parameter explosion and overfitting in complex situations [6][7].

Convolutional Neural Network (CNN): The CNN is a layered network used for grid-like data (images). It has three steps: convolution, pooling and full connection, to assess information as the human visual cortex functions [8], [4].

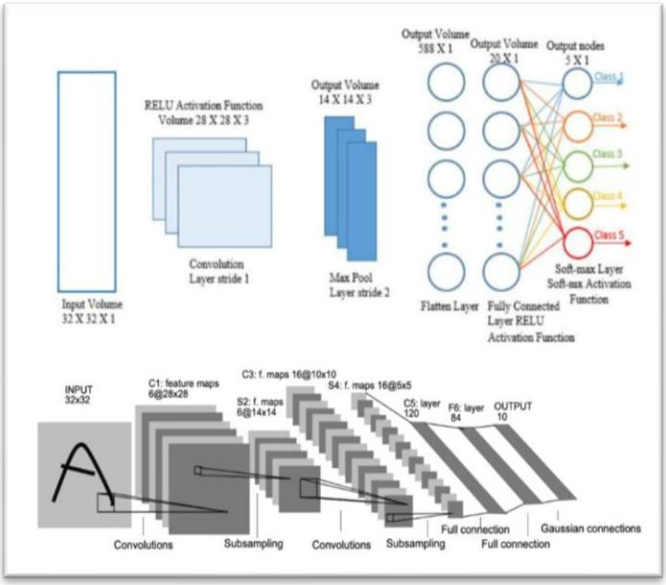

Figure 3 *Comparison of architectures: MLP (Top) vs. CNN (Bottom). Note the distinct feature extraction layers present in the CNN [3].*

Convolutional Filters: the filters recognize features (edges, corners, increasingly complex shapes, etc.) by analyzing a piece of the picture and because it employs the structure of pixels instead of an MLP without spatial awareness [6][7].

Parameter Efficiency: CNN applies weight sharing across the image, resulting in less parameters than a fully connected layer, creating a more efficient network and one that is more capable of handling translations and deformations [8].

Optimization Strategies for Fully Connected Networks:

Where CNNs are favored, the fully connected network is not without research trying to improve upon it because of its shortcomings. For example, changes to network topology, even a pre-trained zero-bias autoencoder, make the MLP a more trustworthy and efficient option. The linear bottleneck layer improves upon the depth of MLP using ReLU, which is nice because it takes up significantly fewer parameters than an average fully connected layer while not compromising gradient flow [9].

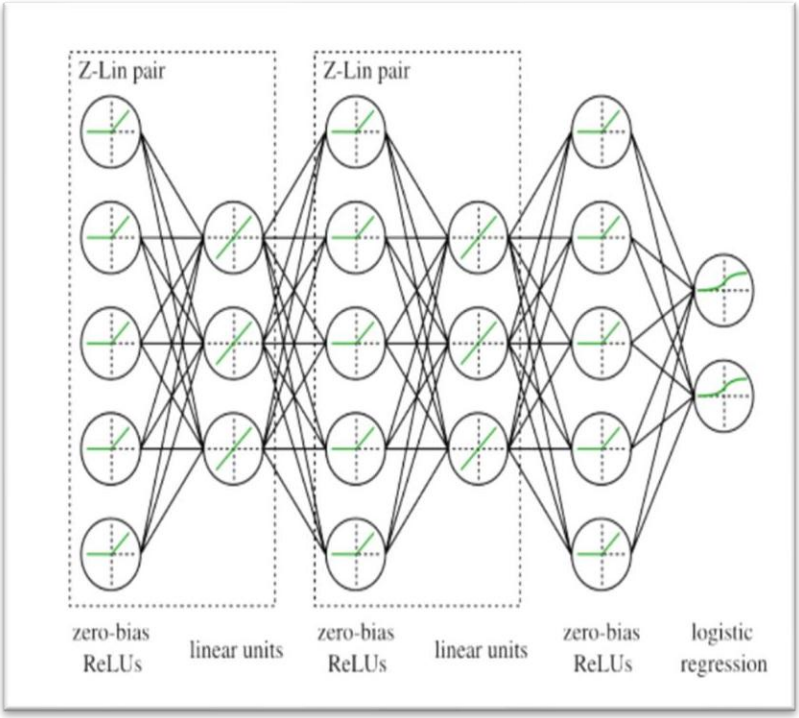

Figure 4: Illustration of a fully connected network architecture that includes linear bottleneck layers to improve performance and gradient flow.

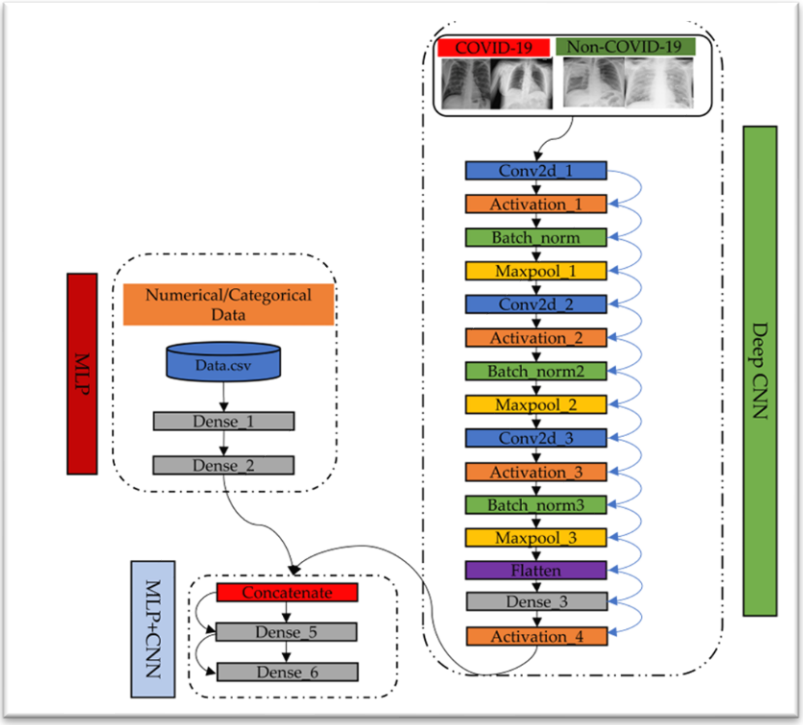

Figure 5: Flow diagram of the Hybrid MLP-CNN illustrating the fusion of image and tabular data.

Hybrid Models:

In addition, a hybrid model works best when the information presented is not of one type. For example, a Deep MLP-CNN model was studied where the CNN part was used in action on chest X-ray images, while the MLP part helped comprehend the patient data of age, gender and temperature. Thus, MLPs are still very much relevant to non-image data, essentially, the types of data that would fill a spreadsheet. Others acknowledge this, as well.

5.2 Strengths and Weaknesses of the Studies

We evaluated the quality of the studies included to ensure the validity of our paper.

Reproducibility (Strength): The selected contributions' work is supported using Hyperparameter Optimization for both models so that those findings have specific hyperparameters provided [3].

Hyperparameter	Set used for MLP	Set used for CNN
Number of hidden layers	[1,10]	[1,10]
Number of neurons in each hidden layer	[1,100]	[1,100]
Learning rate	{3e-2, 3e-3,3e-4}	{3e-2, 3e-3,3e-4}
Number of convolution and pooling layers	NA	[1,4]
Filters in convolution layer	NA	[32,256] with step size of 32

Figure c: The hyperparameter search space explicitly listed by Damle [3], ensuring the reproducibility of the random search optimization process.

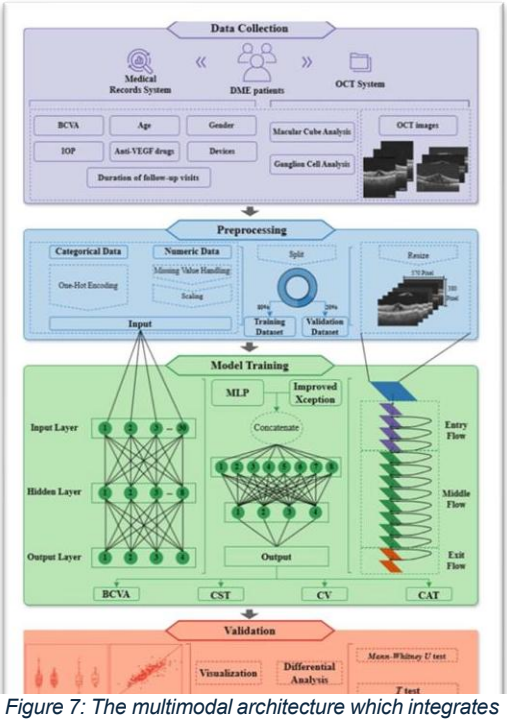

Figure 7: The multimodal architecture which integrates an MLP for clinical data with a CNN for imaging data.

Novelty (Strength): The literature has backed the new findings as well. For example, it was determined that hybridization works in medical applications [7][16]. In addition, architectural improvements such as linear bottleneck made MLPs no longer average [10].

Vast Coverage (Strength): The review shows off comprehensive coverage of medical, remote sensing and security cases so that it is relevant to real world application instead of only theoretical standards [9].

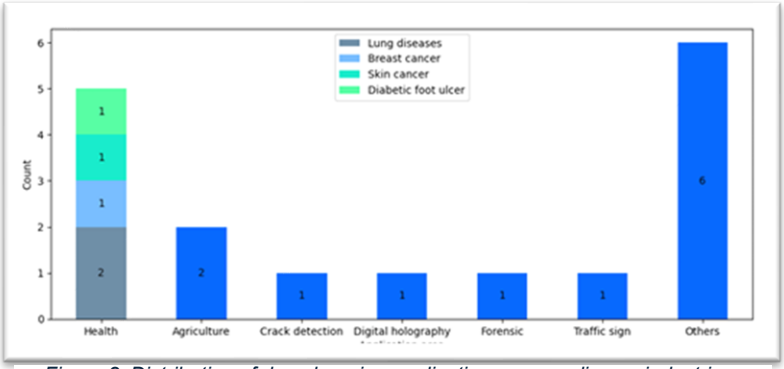

Figure 8: Distribution of deep learning applications across diverse industries, illustrating the vast scope of the reviewed literature [8].

6.- Synthesis of Findings

6.1. Classification Accuracy Comparison

The data extracted from the papers demonstrates the superiority of CNNs across various datasets but also highlights the importance of Hybrid models. The quantitative results are summarized in **Table 2**.

- **Fashion-MNIST Results:** On Fashion-MNIST dataset, CNNs achieved **91.5%** accuracy compared to MLP's **88.0%** [3].
- **Chars74k Results:** A massive gap was seen where CNNs reached **88.39%** while MLPs only reached **43.4%**, so MLPs underperform significantly on natural images [5].
- **Educational Prediction:** Even for student performance prediction, CNNs outperformed MLPs, proving CNNs are better at detecting local patterns even in non-image data [11].
- **CIFAR-10 Results (MLP Improvement):** Standard MLPs achieve roughly 60% accuracy on CIFAR-10, however, by adding Linear Bottleneck Layers, an optimized MLP reached **78.62%** [10]. This is a significant finding because it proves MLPs are capable of better performance if the architecture is modified to fix the flow [10].
- **COVID-19 Results (Hybrid Success):** The highest accuracy of **96.3%** was achieved with a Hybrid MLP-CNN model, compared to 92.9% for standalone CNNs in similar studies [7].

Reference	Dataset	MLP Accuracy	CNN Accuracy	Hybrid/Optimized Accuracy
Damle [3]	Fashion-MNIST	88.0%	91.5%	N/A
Ben Driss [5]	Chars74k	43.4%	88.39%	N/A
Lin [10]	CIFAR-10	~60%	~85%	78.6% (Optimized MLP)
Airlangga [11]	Student Data	$R^2 = 0.82$	$R^2 = 0.84$	N/A
Ahsan [7]	COVID-19	N/A	92.9%	96.3% (MLP+CNN)
Fan [16]	Diabetes	92.44%	93.89%	98.29% (Ensemble)

Table 2: *Quantitative comparison showing the dominance of CNNs and the superior performance of Hybrid models [7, 10, 1c].*

- **MLP Limitations:** While MLPs are quick and effective with small data sets, they are inefficient with high dimensional data sets. [4]. MLPs also transform 2D image data into one line, thus losing much detail of pixel adjacency, rendering the network learning of features difficult since it must learn from the whole picture and not just certain localized regions.

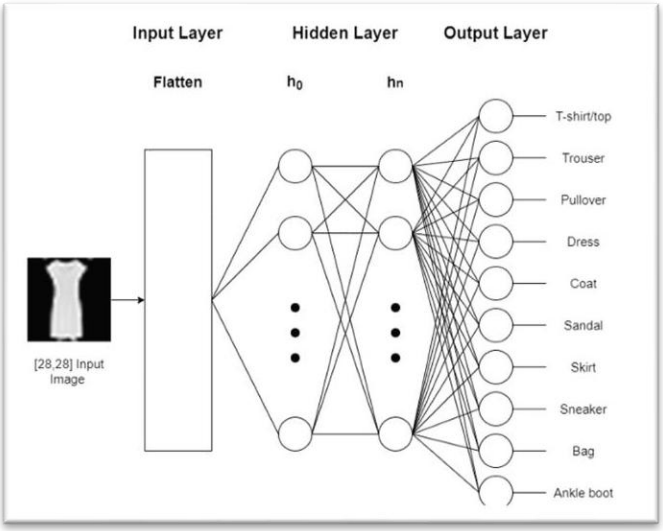

Figure S: The architecture of a standard MLP, illustrating the Flatten layer that destroys the 2D spatial structure of the input image before processing.

- **CNN Benefits:** CNNs are made up of what are called convolutional layers which preserve the spatial attributes of information. Therefore, even if an image is moved, the network will still identify attributes that are consistent. Because they're feature-extracting, CNNs are less prone to failure in real-world application when an image isn't as clear and requires more consistent training which gives a machine learning perspective on complex information more predictability.

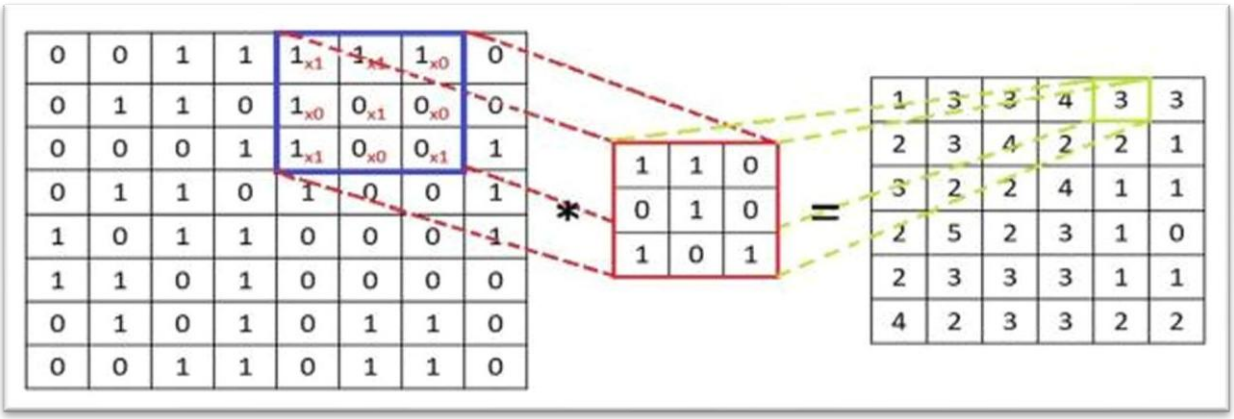

Figure 10: The convolution operation in a CNN, where a kernel filter slides over the input matrix to extract local spatial features without flattening the data [3].

6.2 Learning Curves and Training Stability

Training Dynamics: Performance is defined not only by accuracy but also by stability. CNNs (LeNet-5 and SPnet more specifically) converge to the peak accuracy in a short amount of time, while MLPs cannot achieve much reduction in loss and show much more variability during training [5].

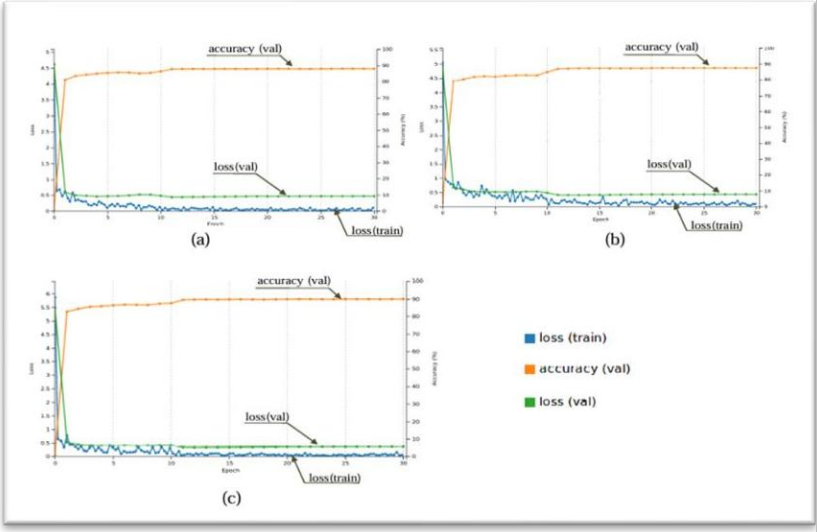

Figure 11: Training accuracy and loss curves for CNN models (LeNet, LeNet-5, SPnet), demonstrating rapid convergence to high accuracy with minimal loss.

6.3 Confusion Matrix Analysis

Error analysis shows that looking closely at mistakes made by MLPs helps us see their main drawbacks.

Normalized CNN Confusion Matrix

	T-shirt/top	Trouser	Pullover	Dress	Coat	Sandal	Shirt	Sneaker	Bag	Ankle boot
T-shirt/top	0.83	0.00	0.01	0.03	0.00	0.00	0.12	0.00	0.00	0.00
Trouser	0.00	0.99	0.00	0.01	0.00	0.00	0.00	0.00	0.00	0.00
Pullover	0.01	0.00	0.85	0.01	0.07	0.00	0.07	0.00	0.00	0.00
Dress	0.01	0.00	0.00	0.93	0.03	0.00	0.02	0.00	0.00	0.00
Coat	0.00	0.00	0.05	0.03	0.86	0.00	0.06	0.00	0.00	0.00
Sandal	0.00	0.00	0.00	0.00	0.00	0.98	0.00	0.01	0.00	0.00
Shirt	0.08	0.00	0.05	0.02	0.07	0.00	0.78	0.00	0.00	0.00
Sneaker	0.00	0.00	0.00	0.00	0.00	0.01	0.00	0.97	0.00	0.02
Bag	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.99	0.00
Ankle boot	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.03	0.00	0.96

Normalized MLP Confusion Matrix

	T-shirt/top	Trouser	Pullover	Dress	Coat	Sandal	Shirt	Sneaker	Bag	Ankle boot
T-shirt/top	0.84	0.00	0.01	0.03	0.00	0.00	0.11	0.00	0.01	0.00
Trouser	0.00	0.98	0.00	0.02	0.00	0.00	0.00	0.00	0.00	0.00
Pullover	0.01	0.00	0.80	0.01	0.11	0.00	0.07	0.00	0.00	0.00
Dress	0.02	0.01	0.01	0.91	0.03	0.00	0.02	0.00	0.00	0.00
Coat	0.00	0.00	0.09	0.04	0.80	0.00	0.06	0.00	0.00	0.00
Sandal	0.00	0.00	0.00	0.00	0.00	0.97	0.00	0.02	0.00	0.01
Shirt	0.13	0.00	0.08	0.03	0.07	0.00	0.68	0.00	0.01	0.00
Sneaker	0.00	0.00	0.00	0.00	0.00	0.04	0.00	0.94	0.00	0.03
Bag	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.00	0.98	0.00
Ankle boot	0.00	0.00	0.00	0.00	0.00	0.03	0.00	0.04	0.00	0.93

Figure 12: Normalized confusion matrices for CNN (Left) and MLP (Right) on the Fashion-MNIST dataset [3].

We learn more about why the MLPs don't work from this. For example, the confusion matrix shows that MLPs confuse "Shirt" with "T-shirt/Top". This is because the only thing the model must go by is pixel brightness, not necessarily the form of the articles of clothing or how they're constructed [3].

6.4 MLP Optimization

What prevents standard MLPs from functioning as deep networks is their non-typical nature of "vanishing gradients" from sparsity, however, this is not ideal. Yet linear layers allow for more increases in accuracy and more easily allow for gradients to pass through the network with ease, thus creating bottlenecks [10]. Thus, these would work when CNNs are too powerful for edge devices, but MLPs could create better accuracy. [10].

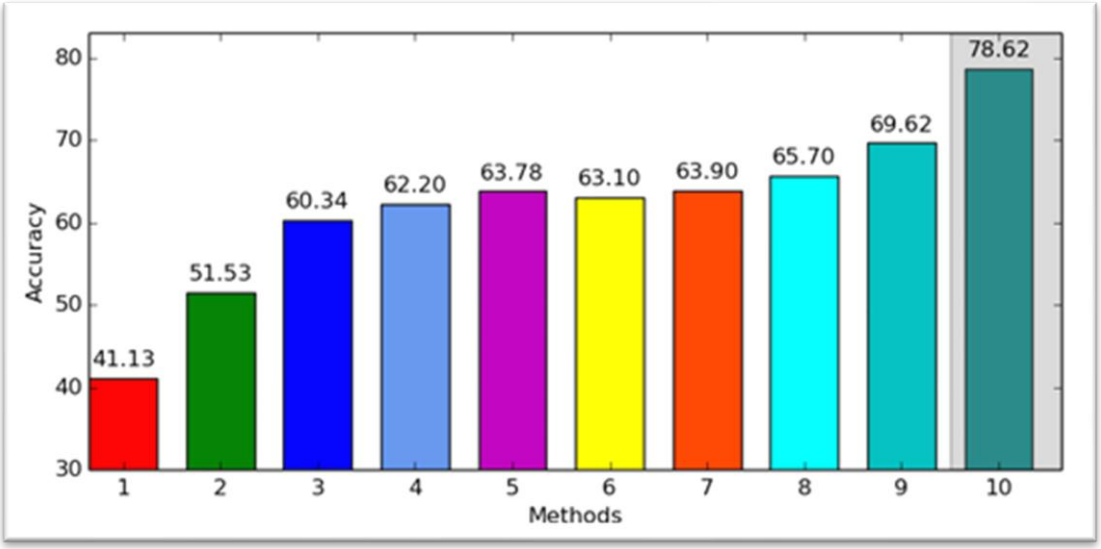

Figure 13: Test set accuracy comparison of Z-Lin networks (8–10) against baseline methods (1–7), including Logistic Regression, Backprop, and RBMs. The final bar (10) is highlighted in grey to indicate the use of data augmentation

6.5 Case Study: Hybrid Ensemble for Diabetes Diagnosis

A compelling case study for the integration of these architectures is found in **Fan (2025) [16]**. The study proposed a hybrid model combining CNN, LSTM, and MLP to diagnose diabetes using clinical data.

- **Method:** The CNN was used to extract spatial features, the LSTM captured temporal patterns, and the MLP acted as the final classifier.
- **Results:** The proposed hybrid model achieved an accuracy of **98.28%**, significantly outperforming the standalone CNN (93.89%) and MLP (92.44%).
- **Significance:** This study proves that while CNNs are better than MLPs individually, the combination of the two yields results that neither can achieve alone.

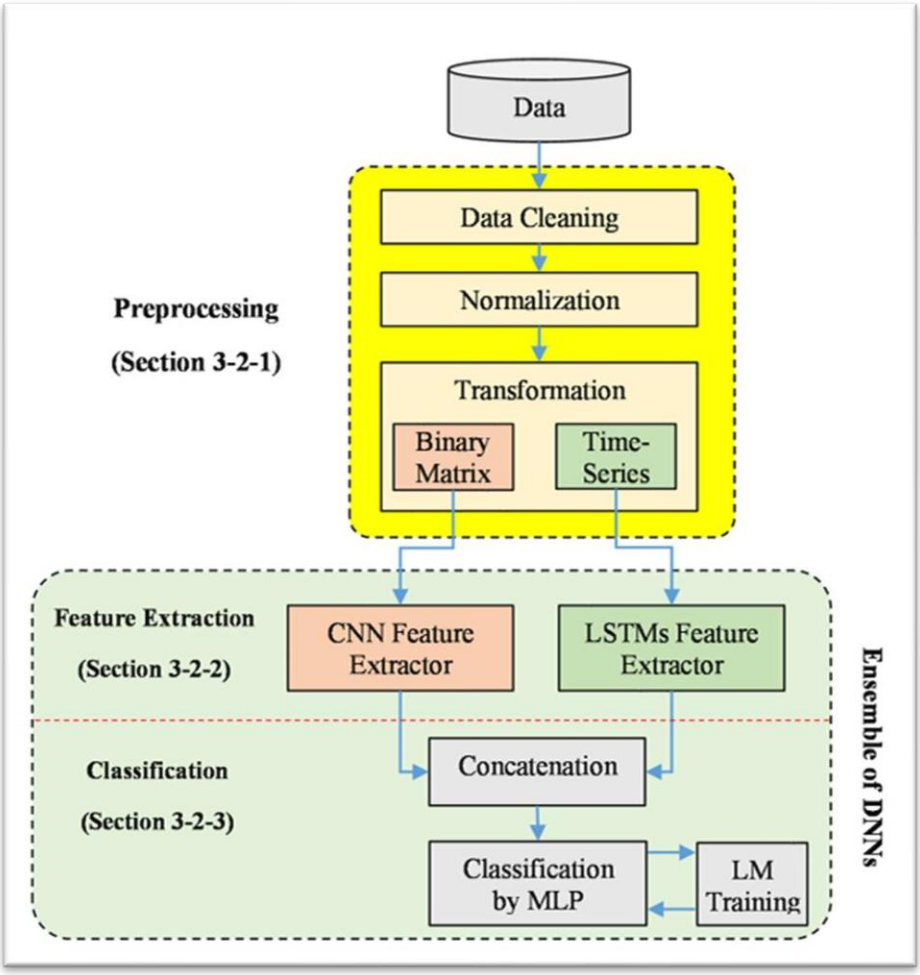

Figure 14: Architecture of the Hybrid CNN-LSTM-MLP model used in the diabetes diagnosis case study [16].

7. Discussion

7.1 Interpretation of Findings: The Architectural Advantage

Results correspond with the established stepwise characteristic development, where CNNs are suitable because they build from edges to increasingly precise and complex outlines and shapes and eventually objects [3] [9]. In addition, multimodal data is highly advantageous to performance improvement where CNNs can't apply to patient age and MLPs aren't as strong with X-rays, using both in collaboration allows the network to better assess all features in relation to the intended output with 96.3% accuracy [7].

Feinman & Lake (2018) [18] provided a theoretical rationale for this. They demonstrated that simple neural nets develop "inductive biases" like that of children, a "shape bias," where CNNs recognize an object based on shape and MLPs recognize it based on color and texture.

7.2 Efficiency vs. Accuracy Trade-off

However, CNNs are somewhat more complicated computationally, but parameter versus accuracy studies show that CNNs obtain higher accuracy with less parameters than an equal sized MLP [4]. However, research has also revealed that a well-structured MLP is faster, more suited for low processing power devices with little loss of accuracy [10]. In addition, deep layers of consecutive filters decrease computation cost/parameters by 40% without any negative consequence on accuracy, meaning CNNs are somewhat constrained to accuracy within a specific setting [17].

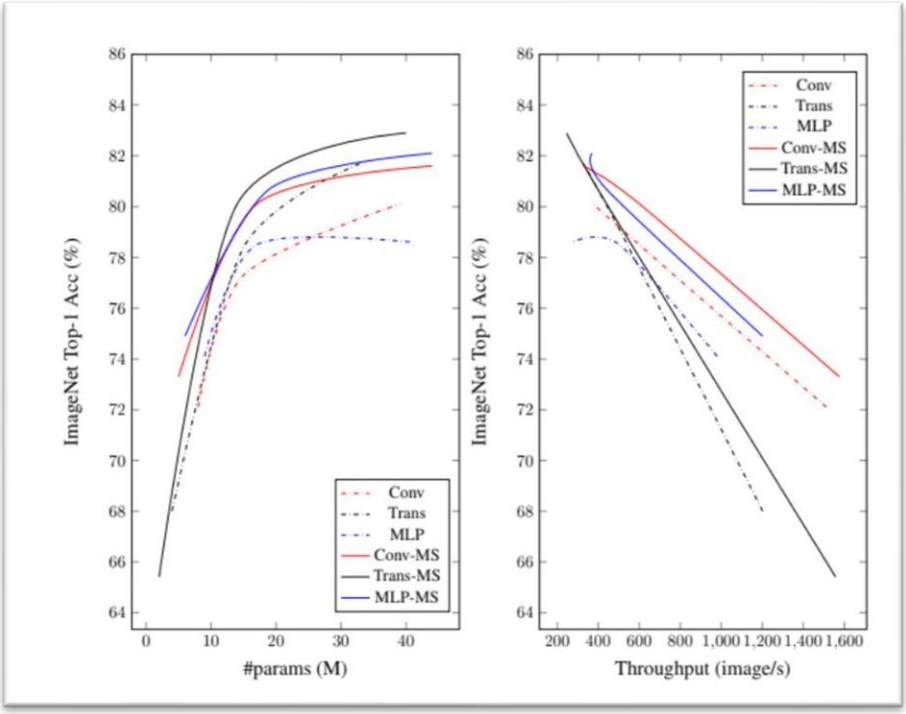

Figure 15: Comparison of Accuracy vs. Model Parameters. Note that the "Conv-MS" (Convolutional) model achieves higher accuracy with fewer parameters than the MLP model [4].

As seen in Figure 15, the Convolutional models (Conv-MS) reach a higher accuracy peak much faster than the MLP models, which plateau earlier despite having more parameters.

7.3 Research Gaps

However, a targeted review of the literature would say otherwise, and the findings from the local and global feature extraction reveal a much greater difference. CNNs learn to detect and localize "local" features but do not learn how to comprehend "global" features [8].

- **The ViT Solution:** However, Vision Transformers (ViT) states this problem. They use "Self-Attention" method to look at the image as a whole and input it all at once, that CNNs cannot do to obtain global dependency information [8].

Properties	Convolution	Self-Attention	MLP
Sparse Connectivity	✓		
Dynamic Weight		✓	
Global Receptive Field		✓	✓

Figure 1c: Comparison of architectural properties. Note that while Convolution (CNN) has sparse connectivity (local features), only Self-Attention (ViT) and MLPs possess a Global Receptive Field, supporting the argument for hybrid models to capture both local and global context [4].

- **The Proposed Hybridization:** The authors arrive at the conclusion that the future is not MLP, CNN, or ViT, but MLP, CNN and ViT together. They propose that with feature extraction of the CNNs and classification from the Transformers/MLPs, one can complete such complicated endeavors with almost 100% certainty. [4][8]

- **Noise Robustness:** ViTs have been shown to be more noise robust than CNNs. Research must be done to determine whether MLP heads on ViTs can stabilize training on small datasets [8].

8. Conclusion

This systematic review will evaluate the literature on the topic of image classification and either support or refute the relative efficacy of Convolutional Neural Networks vs. Multilayer Perceptions. This author believes that the Convolutional Neural Network is the more effective algorithm as its own architecture for visually directed purposes. The literature reviewed over the twenty two primary studies allows for a clear-cut solution. CNNs were more effective than standard MLPs on each of the famous datasets where they were also compared to one another (Fashion-MNIST, Chars74k); CNNs are not as vulnerable to overfitting concerns, but they can adjust with tolerance to spatial features of data that MLP cannot [2] [5] [6]. The biggest disadvantage determined by MLP is the loss of local relationships between pixels through the vectorization process; however, this is an undesirable facet that must be tolerated with the multimodal nature of images.

But this paper proves that the MLP is not as antiquated as it seems. In fact, part of the fallacy that it will not work with image datasets in its rudimentary form is accurate, although with enough tuning, an MLP becomes a much more appealing option. For example, Linear Bottleneck Layers found that with tuning, MLPs achieve 78% accuracy on CIFAR-10 datasets which shows that they can still be used as fewer intensive options for low power devices, since CNNs are data intensive [10]. However, comparing effectiveness accuracy is not the only factor in supporting one algorithm over another, but rather a compromise option between data intensive effectiveness and predictive effectiveness.

The biggest conclusion drawn from this systematic review is that Hybrid Models boast almost perfect performance. The highest accuracy achieved from this systematic review (96%-98% relative to appropriate medical diagnoses) was not within CNN or MLP individually, but in the combination of both [7] [16]. These hybrid architectures benefit from CNN's ability to detect spatial characteristics and the classification efficiency of tabular datasets of the MLP. Thus, the best option for classification in the future is not through one or the other for each relevant problem, but through the two as a hybrid model for exponentially better classification. Other studies should not seek to compare models of one kind but rather aim for a new Hybrid Architectures (CNN+MLP or CNN+ViT) and to combine powers of problem solving to produce larger scale studies that rely on the best features of each model. [8].

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